

UNIVERSITY OF TWENTE.

THE SUSTAINABILITY OF DIGITAL TRACE DATA FOR BEHAVIORAL RESEARCH

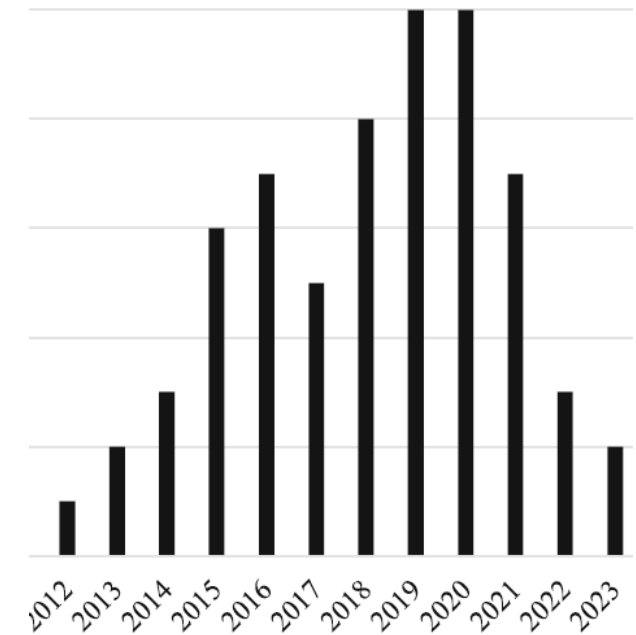
Context

Organizational behavior of energy-efficient computing

More AI, ML, Cloud in information systems research:

→ growing interest for computational approaches that require energy.

Focus on **digital trace data** research.



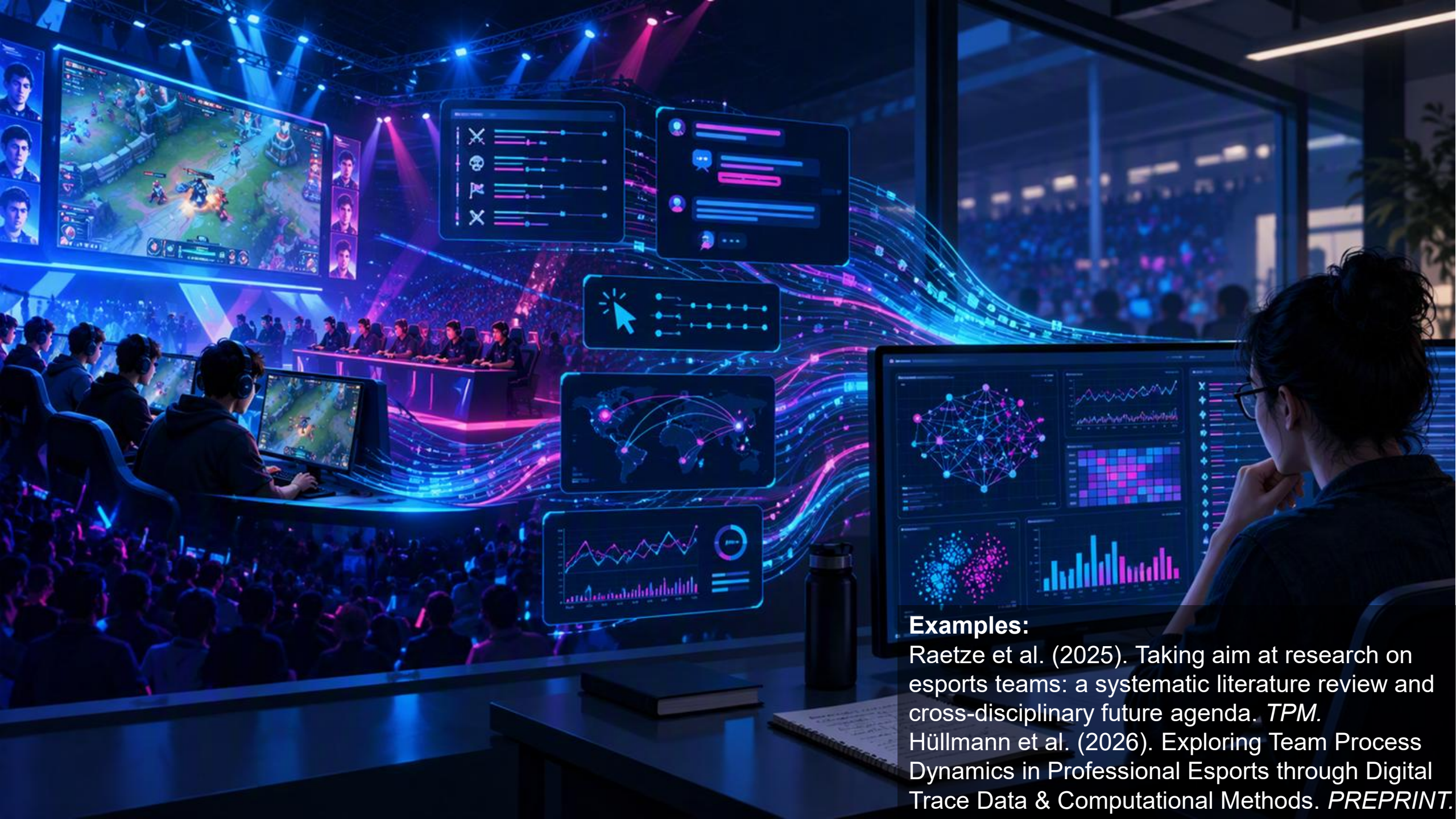
Papers on digital trace
data per year

(Hüllmann, 2026)



Examples:

Cao et al. (2021). Large scale analysis of multitasking behavior during remote meetings. *CHI*.
Hüllmann & Krebber (2020). Identifying Temporal Rhythms using Email Traces. *AMCIS*.



Examples:

Raetz et al. (2025). Taking aim at research on esports teams: a systematic literature review and cross-disciplinary future agenda. *TPM*.
Hüllmann et al. (2026). Exploring Team Process Dynamics in Professional Esports through Digital Trace Data & Computational Methods. *PREPRINT*.

Digital Trace Data Research Process

1. People use IT and generate **low-level** behavioral trace data as a by-product.
2. ETL to curate a data set with **meaningful variables** for behavioral research.
3. Run statistical/computational analyses.

Energy to operate and use IT.

Energy to collect and store data, and run computational procedures (e.g., machine learning classifiers).

Energy to store data and run analyses.

Example: Counter-Strike Tactics

Counter-Strike:

- Low-level trace data from replays
- Use **(Graph) Neural Networks** to estimate tactics (e.g., „Execute B“)
- Use tactics in follow-up regressions

equippedWeapon	positionX	positionY	positionZ
Knife	-1890.52783203125	-1926.513916015625	-272.6513977050781
USP-S	-1647.7130126953125	-1751.55078125	-267.00885009765625
Knife	-1917.4442138671875	-1988.952392578125	-275.78729248046875
Smoke Grenade	-1862.55419921875	-1852.1279296875	-269.392333984375
Knife	-1588.443603515625	-2100.6005859375	-261.7655334472656
Knife	1194.6082763671875	34.34914779663086	-163.96875
Knife	1257.71435546875	-129.3793487548828	-167.96875
Knife	1352.462158203125	32.7823486328125	-167.96875
Smoke Grenade	1349.2615966796875	-347.0602722167969	-167.96875
Smoke Grenade	1291.2513427734375	-64.99573516845703	-129.70213317871094
Knife	-1738.5645751953125	-1751.976318359375	-265.7339782714844
Knife	-1733.7813720703125	-1887.3614501953125	-269.3951110839844
Knife	-1401.8992919921875	-2242.90698	



Dilemma

“Secondary data usage enables a resource-saving use of public funds, and optimal knowledge gain through cumulative research.”

–German Academy of Sociology (2020)

Reusing digital trace data is hard:

- ETL'ing from proprietary systems running in different contexts & configurations
- High researcher degrees of freedom and low paradigm
- Hard to reuse intermediate results for adjacent Q's (except direct replications)
(Hüllmann, 2026)

Energy-Efficiency through Collective Behavior

(first ideas)

Steps 1 and 2:

- Modularize, standardize, and build an interfaceable centralized platform (Grisold et al., 2023)
- Develop more guidelines for how to ETL digital trace data
- Enhance computational reproducibility (Reiss et al., 2026)

Steps 2 and 3:

- Reusable, transferable, intermediate ML models?

Comment

<https://doi.org/10.1038/s41562-026-02570-w>

Improving computational reproducibility in the social sciences

Michael V. Reiss, Susanne J. Adler, Christopher Barrie, Jana B. Berkessel, Johannes Breuer, Abel Brodeur, Garret Christensen, Tobias Dienlin, Jeremy Freese, Ben Greiner, Mario Haim, Tom E. Hardwicke, Magnus Johannesson, Gary King, Frauke Kreuter, Christopher Lucas, Ana Martinovici, Christophe Pérignon, Marko Sarstedt, Felix D. Schönbrodt, Oleg Urminsky, Wouter van Atteveldt, Simine Vazire, Lars Vilhuber, Florian von Wangenheim, Eric-Jan Wagenmakers, Nicolas Pröllochs, Claire Robertson, Felix Holzmeister, Stefan Feuerriegel & Hauke Roggenhamp

 Check for updates

Code underlying published findings in the social sciences often fails to reproduce reported results when others re-run it on the original data. This Comment proposes four recommendations to strengthen computational reproducibility, facilitating the trustworthiness and reliability of research.